

Kneser graphs $K(2k + n, k)$ with $n > 0$ are not CFON* 1-colorable:
the general proof

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1 Efficient open dominating sets and CFON* 1-colorings

An efficient open dominating set of a Kneser graph K is a subset T of the vertices of K such that every vertex of K has exactly one neighbor in T .

Note that an efficient open dominating set is analogous to a CFON* 1-coloring: if T is the set of all colored vertices, then the coloring is valid exactly when every vertex has exactly one neighbor in T (since we only have one color, we can treat all colored vertices as the same).

Assign each vertex v of K a variable:

$$x_v = \begin{cases} 1, & v \in T \\ 0, & v \notin T \end{cases}$$

Thus, for every vertex v , the variables of the vertices in its open neighborhood $N(v)$ sum to 1.

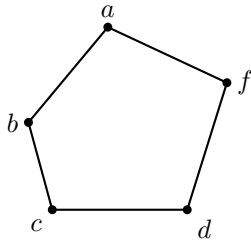
2 The matrix equation

So for every vertex v , summing over the vertices u in $N(v)$:

$$\sum_{u \in N(v)} x_u = \underbrace{0 + 0 + 0 + \cdots}_{|N(v)|-1} + \underbrace{1}_{\text{the colored vertex}} = 1$$

Let A be a matrix in which each row and each column represents a vertex. Let an element equal 1 if the vertices of its row and column are adjacent, and 0 otherwise.

Example:



$$\begin{matrix} & a & b & c & d & f \\ \begin{matrix} a \\ b \\ c \\ d \\ f \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Let x be a $|V(K)| \times 1$ matrix (a vector) in which each element represents a vertex. Let an element equal 1 if the corresponding vertex is colored, and 0 otherwise.

$$Ax = A \begin{bmatrix} x_a \\ x_b \\ \vdots \\ x_f \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \leftarrow \begin{array}{l} \text{the number of colored vertices in the} \\ \text{neighborhood of the corresponding} \\ \text{vertex} \end{array}$$

The all-ones vector on the right is written $\mathbf{1}$ from here on.

3 The general case $K(2k + n, k)$

We generalize to $K(2k + n, k)$ with $k \geq 1$ and $n > 0$.

The graph is $\binom{k+n}{k}$ -regular.

Let M be its adjacency matrix: each row and each column corresponds to a vertex, and an element equals 1 if the vertices of its row and column are adjacent, and 0 otherwise. Each row of M has $\binom{k+n}{k}$ 1's.

Consider the equation $Mq = \mathbf{1}$, and suppose every element of q is the same constant g :

$$q = \begin{bmatrix} g \\ g \\ \vdots \\ g \end{bmatrix}$$

Then each row gives

$$\left(\frac{(k+n)!}{k! n!} \right) g = 1, \quad g = \frac{k! n!}{(k+n)!}$$

The eigenvalues of M are

$$\lambda_j = (-1)^j \binom{k+n-j}{k-j}, \quad j = 0, 1, 2, \dots, k$$

Their magnitudes are never 0, because $k+n-j \geq k-j \geq 0$ for every j , which makes every $\binom{k+n-j}{k-j}$ at least 1. Thus every eigenvalue of M is non-zero, making M invertible.

Now suppose there were some vector $w \in \mathbb{N}_0^{|V(K)|}$ with $Mw = \mathbf{1}$, so that w meets the CFON* coloring restrictions.

Then:

$$\begin{aligned} Mw &= Mq = \mathbf{1} \\ M^{-1}Mw &= M^{-1}Mq && \text{(since } M \text{ is invertible)} \\ w &= q \end{aligned}$$

But q is fractional: n is strictly greater than 0, which makes $\binom{k+n}{k} \geq k+1 \geq 2$, and so $0 < g < 1$. This contradicts the definition of w . Thus w cannot exist, and q is the unique solution.

Since q is the unique solution and all its elements are fractional, while each x_v can only equal 0 or 1 depending on whether v is in T , no efficient open dominating set exists, and thus $K(2k+n, k)$ is not CFON* 1-colorable.